

点屏

因为ssd20x clk的限制，点panel前先确认panel的clk是否支持，ssd20x clk的spec如下：

TTL输出CLK支持范围9Mhz-75Mhz,

计算方法: $htotal * vtotal * fps$

MIPI DSI输出CLK支持范围100Mbps/lane - 1.5Gbps/lane

计算方法: $H_Total * V_Total * FPS * BitsPerPixel / lane \ number$

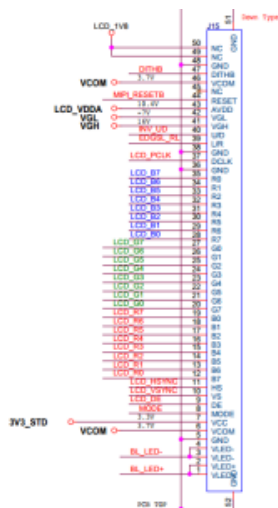
BitsPerPixel=24 (RGB888) / 18 (RGB666) / 16 (RGB565)

Panel相关pin脚的配置

首先跟硬件确认好是用那组pin脚(最好是使用下文中所列的pin, 这些pin我们都有验证过), 然后在kernel的dts里面切到对应的mode,

TTL

以公版为例，用的是PAD_TTL0~PAD_TTL27这组：



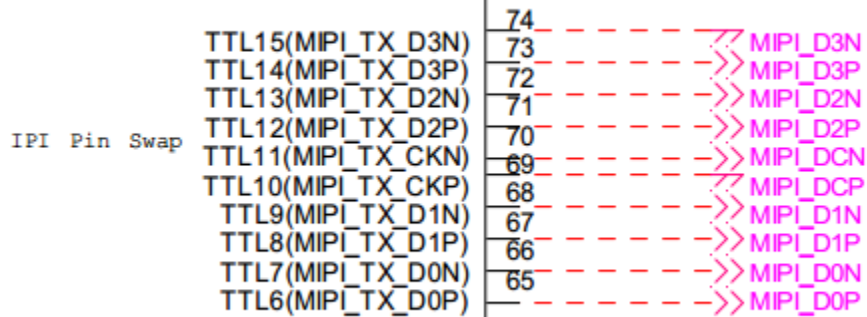
PAD_TTL0	R171	0R	LCD_R0			
PAD_TTL1	R169	0R	LCD_R1	<<	LCD_R0	9
PAD_TTL2	R173	0R	LCD_R2	<<	LCD_R1	9
PAD_TTL3	R170	0R	LCD_R3	<<	LCD_R2	9
PAD_TTL4	R204	0R	LCD_R4	<<	LCD_R3	9
PAD_TTL5	R202	0R	LCD_R5	<<	LCD_R4	9
PAD_TTL6	R175	0R	LCD_R6	<<	LCD_R5	9
PAD_TTL7	R172	0R	LCD_R7	<<	LCD_R6	9
				<<	LCD_R7	9
PAD_TTL8	R176	0R	LCD_G0			
PAD_TTL9	R174	0R	LCD_G1	<<	LCD_G0	9
PAD_TTL10	R179	0R	LCD_G2	<<	LCD_G1	9
PAD_TTL11	R177	0R	LCD_G3	<<	LCD_G2	9
PAD_TTL12	R181	0R	LCD_G4	<<	LCD_G3	9
PAD_TTL13	R178	0R	LCD_G5	<<	LCD_G4	9
PAD_TTL14	R182	0R	LCD_G6	<<	LCD_G5	9
PAD_TTL15	R180	0R	LCD_G7	<<	LCD_G6	9
				<<	LCD_G7	9
PAD_TTL16	R186	0R	LCD_B0			
PAD_TTL17	R196	0R	LCD_B1	<<	LCD_B0	9
PAD_TTL18	R189	0R	LCD_B2	<<	LCD_B1	9
PAD_TTL19	R197	0R	LCD_B3	<<	LCD_B2	9
PAD_TTL20	R201	0R	LCD_B4	<<	LCD_B3	9
PAD_TTL21	R198	0R	LCD_B5	<<	LCD_B4	9
PAD_TTL22	R203	0R	LCD_B6	<<	LCD_B5	9
PAD_TTL23	R200	0R	LCD_B7	<<	LCD_B6	9
				<<	LCD_B7	9
PAD_TTL24	R150	0R	LCD_PCLK			
PAD_TTL25	R151	0R	LCD_HSYNC	<<	LCD_PCLK	9
PAD_TTL26	R152	0R	LCD_VSYNC	<<	LCD_HSYNC	9
PAD_TTL27	R153	0R	LCD_DE	<<	LCD_VSYNC	9
				<<	LCD_DE	9

在infinity2m-ssc011a-s01a-padmux-display.dtsi中配置成PINMUX FOR TTL MODE 1

<PAD_FUART_RTS	PINMUX_FOR_GPIO_MODE	MDRV_FUSE_CPFUSE_VID1>
<PAD_TTL0	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT00 >
<PAD_TTL1	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT01 >
<PAD_TTL2	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT02 >
<PAD_TTL3	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT03 >
<PAD_TTL4	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT04 >
<PAD_TTL5	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT05 >
<PAD_TTL6	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT06 >
<PAD_TTL7	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT07 >
<PAD_TTL8	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT08 >
<PAD_TTL9	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT09 >
<PAD_TTL10	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT10 >
<PAD_TTL11	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT11 >
<PAD_TTL12	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT12 >
<PAD_TTL13	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT13 >
<PAD_TTL14	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT14 >
<PAD_TTL15	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT15 >
<PAD_TTL16	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT16 >
<PAD_TTL17	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT17 >
<PAD_TTL18	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT18 >
<PAD_TTL19	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT19 >
<PAD_TTL20	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT20 >
<PAD_TTL21	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT21 >
<PAD_TTL22	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT22 >
<PAD_TTL23	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DOUT23 >
<PAD_TTL24	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_CLK >
<PAD_TTL25	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_HSYNC >
<PAD_TTL26	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_VSYNC >
<PAD_TTL27	PINMUX_FOR_TTL_MODE_1	MDRV_FUSE_TTL_DE >
<PAD_UART0_RX	PINMUX_FOR_UART0_MODE_1	MDRV_FUSE_UART0_RX >
<PAD_UART0_TX	PINMUX_FOR_UART0_MODE_1	MDRV_FUSE_UART0_TX >

MIPI

如下图所示,用的是PAD TTL6~PAD TTL15:

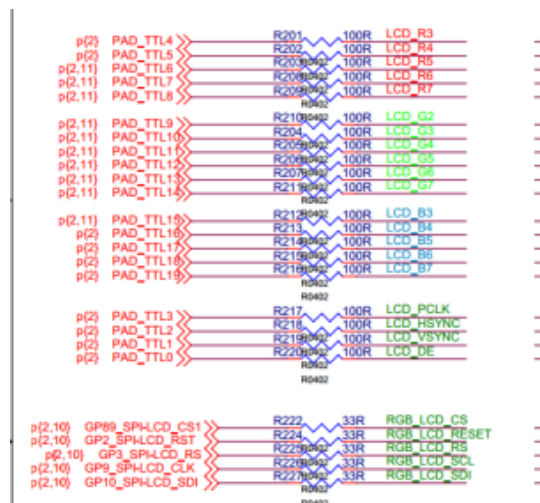


所以在padmux里面配置成PINMUX_FOR_TX_MIPI_MODE_1 mode:

```
//<PAD_GPIO17>,
//<PAD_FUART_RX>,
//<PAD_FUART_TX>.....PINMUX_FOR_GPIO_MODE.....MDRV_PUSE_UTMI_POWER>,
<PAD_FUART_CTS>.....PINMUX_FOR_GPIO_MODE.....MDRV_PUSE_CPUFREQ_VID0>,
<PAD_FUART_RTS>.....PINMUX_FOR_GPIO_MODE.....MDRV_PUSE_CPUFREQ_VID1>,
//<PAD_TTL0>.....PINMUX_FOR_SDIO_MODE_1.....MDRV_PUSE_SDIO0_FWR>,
<PAD_TTL6>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_P_CH0>,
<PAD_TTL7>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_N_CH0>,
<PAD_TTL8>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_P_CH1>,
<PAD_TTL9>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_N_CH1>,
<PAD_TTL10>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_P_CH2>,
<PAD_TTL11>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_N_CH2>,
<PAD_TTL12>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_P_CH3>,
<PAD_TTL13>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_N_CH3>,
<PAD_TTL14>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_P_CH4>,
<PAD_TTL15>.....PINMUX_FOR_TX_MIPI_MODE_1.....MDRV_PUSE_TX_MIPI_N_CH4>,
<PAD_UART0_RX>.....PINMUX_FOR_UART0_MODE_1.....MDRV_PUSE_UART0_RX>,
<PAD_UART0_TX>.....PINMUX_FOR_UART0_MODE_1.....MDRV_PUSE_UART0_TX>,
//<PAD_UART1_RX>,
//<PAD_UART1_TX>,
//<PAD_SD_CLK>.....PINMUX_FOR_I2S_MODE_3.....MDRV_PUSE_I2S_BCK>,
<PAD_SD_CLK>.....PINMUX_FOR_I2S_MODE_3.....MDRV_PUSE_I2S_BCK>
```

RMI1

如下图所示，用的是PAD_TTL0~PAD_TTL19:



所以在padmux里面配置成PINMUX_FOR_TTL_MODE_10:

```

.....<PAD_FUART_CTS.....PINMUX_FOR_I2C1_MODE_3.....MDRV_PUSE_I2C1_SCL>,
.....<PAD_FUART_RTS.....PINMUX_FOR_I2C1_MODE_3.....MDRV_PUSE_I2C1_SDA>,
.....<PAD_TTL0.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DE>,
.....<PAD_TTL1.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_VSYNC>,
.....<PAD_TTL2.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_HSYNC>,
.....<PAD_TTL3.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_CLK>,
.....<PAD_TTL4.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT00>,
.....<PAD_TTL5.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT01>,
.....<PAD_TTL6.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT02>,
.....<PAD_TTL7.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT03>,
.....<PAD_TTL8.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT04>,
.....<PAD_TTL9.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT05>,
.....<PAD_TTL10.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT06>,
.....<PAD_TTL11.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT07>,
.....<PAD_TTL12.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT08>,
.....<PAD_TTL13.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT09>,
.....<PAD_TTL14.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT10>,
.....<PAD_TTL15.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT11>,
.....<PAD_TTL16.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT12>,
.....<PAD_TTL17.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT13>,
.....<PAD_TTL18.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT14>,
.....<PAD_TTL19.....PINMUX_FOR_TTL_MODE_10.....MDRV_PUSE_TTL_DOUT15>,
.....<PAD_TTL20.....PINMUX_FOR_GPIO_MODE.....MDRV_PUSE_TTL_DOUT20>,
.....<PAD_TTL21.....PINMUX_FOR_ETH1_MODE_5.....MDRV_PUSE_NA>,
.....<PAD_TTL22.....PINMUX_FOR_ETH1_MODE_5.....MDRV_PUSE_NA>,
.....<PAD_TTL23.....PINMUX_FOR_ETH1_MODE_5.....MDRV_PUSE_NA>,
.....<PAD_TTL24.....PINMUX_FOR_ETH1_MODE_5.....MDRV_PUSE_NA>,

```

TTL panel

首先找一个已经配好的相同分辨率的ttl 屏参，然后根据panel spec填好以下参数：

u16HSyncWidth & u16HSyncBackPorch & u16VSyncWidth & u16VSyncBackPorch

对应panel spec的数据如下：

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Horizontal Display Area	thd	-	1024	-	DCLK	
DCLK frequency	fck	44.9	51.2	63	MHz	
One horizontal line	th	1200	1344	1400	DCLK	
HS pulse width	thpw	1	-	140	DCLK	
HS Blanking	thb	160	160	160	DCLK	
HS Front Porch	thfp	16	160	216	DCLK	

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Vertical display Area	tvd	-	600	-	TH	
VS period time	tv	624	635	750	TH	
VS pulse width	tvpw	1	-	20	TH	
VS Blanking	tvb	23	23	23	TH	
VS Front Porch	tvfp	1	12	127	TH	

u16HStart = u16HSyncWidth + u16HSyncBackPorch

u16VStart = u16VSyncWidth + u16VSyncBackPorch

u16Width & u16Height对应panel spec数据如下：

6.2 Timing Conditions

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Horizontal Display Area	thd	-	1024	-	DCLK	
DCLK frequency	fck	44.9	51.2	63	MHz	
One horizontal line	th	1200	1344	1400	DCLK	
HS pulse width	thpw	1	-	140	DCLK	
HS Blanking	thb	160	160	160	DCLK	
HS Front Porch	thfp	16	160	216	DCLK	

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Vertical display Area	tvd	-	600	-	TH	
VS period time	tv	624	635	750	TH	
VS pulse width	tvpw	1	-	20	TH	
VS Blanking	tvb	23	23	23	TH	
VS Front Porch	tvfp	1	12	127	TH	

$$u16HTotal = u16Width + u16HSyncWidth + u16HSyncBackPorch + HSyncFrontPorch$$

$$u16VTotal = u16Height + u16VSyncWidth + u16VSyncBackPorch + VSyncFrontPorch$$

对应panel spec数据如下：

6.2 Timing Conditions

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Horizontal Display Area	thd	-	1024	-	DCLK	
DCLK frequency	fck	44.9	51.2	63	MHz	
One horizontal line	th	1200	1344	1400	DCLK	
HS pulse width	thpw	1	-	140	DCLK	
HS Blanking	thb	160	160	160	DCLK	
HS Front Porch	thfp	16	160	216	DCLK	

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Vertical display Area	tvd	-	600	-	TH	
VS period time	tv	624	635	750	TH	
VS pulse width	tvpw	1	-	20	TH	
VS Blanking	tvb	23	23	23	TH	
VS Front Porch	tvfp	1	12	127	TH	

$$u16DCLK = u16HTotal * u16VTotal * frequency(60)/1000*1000$$

如下三个参数分别对应R G B 三个channel的数据是否要做swap

$$u8SwapOdd_RG \ \& \ u8SwapEven_RG \ \& \ u8SwapOdd_GB$$

对应channel选择的情况如下：

0 default(不做swap)

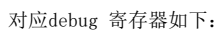
1 select R

2 select G

3 select B

$u8SwapEven_GB$ 表示rgb数据线的高低位是否需要做swap

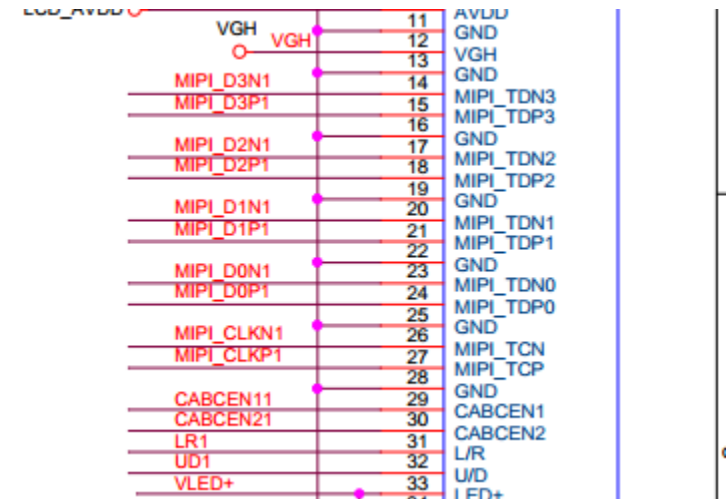
eg: 如demo板原理图如下，R和B以及高低位反了，所以屏参的配置为3,2,1,1

1129

	7e	[bit0-bit5] rgb swap [bit0-bit1] swap for B channel [bit2-bit3] swap for G channel [bit4-bit5] swap for R channel [bit6-bit7] rgb mode 00:rgb 888 01:rgb 666 10:rgb 565-1 11:rgb 565-2 [bit8] MSB/LSB swap
--	----	---

MI_PANEL_ChannelSwapType_e eCh0 eCh1 eCh2 eCh3 eCh4

eg: demo板原理图和配置如下



(MI_PANEL_ChannelSwapType_e)2,
(MI_PANEL_ChannelSwapType_e)4,
(MI_PANEL_ChannelSwapType_e)3,
(MI_PANEL_ChannelSwapType_e)1,
(MI_PANEL_ChannelSwapType_e)0,

MI_PANEL_MipiDsiConfig_t需要重点关注的参数如下:

u16Hactive u16Hpw u16Hbp u16Hfp u16Vactive u16Vpw u16Vbp u16Vfp

这几个参数对应的是panel spec中的 :

u16Width u16HSyncWidth u16HSyncBackPorch HSyncFrontPorch

u16Height u16VSyncWidth u16VSyncBackPorch VSyncFrontPorch

这几个参数需要保持和MI_PANEL_ParamConfig_t一致

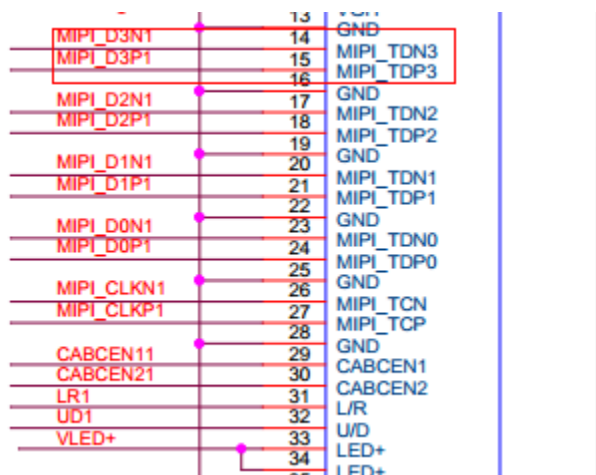
enLaneNum 对应mipi 几line mode根据实际情况配置

pu8CmdBuf u32CmdBufSize 对应mipi的初始化cmd,对应格式如下:

```
regw(0x01); //Reset
Delay 30ms
regw(0x80, 0x58);
regw(0x81, 0x47);
→
0x01, 0, 0x0, //0x01: 寄存器, 0: 参数个数, 0x0:写寄存器的值
FLAG_DELAY, FLAG_DELAY, 30,
0x80, 1, 0x58,
0x81, 1, 0x47,
```

u8PolCh0 u8PolCh1 u8PolCh2 u8PolCh3 u8PolCh4 对应mipi四组data line一组clk line是否需要反转

eg: 如下原理图就不需要反转



如果出现clk&data都正常，但是panel不亮可以微调BK1033_44[15:7]，默认是9，可以左右6 7 8 9 a b c d e f 微调，

这个值是clk和data之间的时间差，对应mipi屏参的最后一个：

```
MI_PANEL_MipiDsiConfig_t stMipiDsiConfig =
{
    //HsTrail HsPrpr HsZero ClkHsPrpr ClkHsExit ClkTrail ClkZero ClkHsPost DaHsExit Cont
    0x05, 0x03, 0x05, 0x0A, 0x0E, 0x03, 0x0B, 0x0A, 0x05, 0x00,
    //Lpx TaGet TaSure TaGo
    0x10, 0x1a, 0x16, 0x32,

    //Hac Hpw Hbp Hfp Vac Vpw Vbp Vfp Blip Fps
    800, 20, 20, 20, 1280, 4, 10, 30, 0, 60,

    E_MI_PNL_MIPI_DSI_LANE_4, // MIPnlMipiDsiLaneMode_e enLaneNum;
    E_MI_PNL_MIPI_DSI_RGB888, // MIPnlMipiDsiFormat_e enFormat;
    E_MI_PNL_MIPI_DSI_SYNC_PULSE, // MIPnlMipiDsiCtrlMode_e enCtrl;

    JD9365_CMD,
    sizeof(JD9365_CMD),
    1, 0x01AF, 0x01B9, 0x80D2, 9,
};
```

SPI panel

- kernel需要enable如下两个config打开spi 驱动：
CONFIG_SPI_SPIDEV=y
CONFIG_MS_SPI_INFINITY=y
- dts配置好spi相关pin，kernel\arch\arm\boot\dts\infinity2m.dtsi

```
spi: spi {
    compatible = "ssstar_spi";
    io_phy_addr = <0x1f000000>;
    banks = <0x1110>, <0x1111>, <0x1038>, <0x101E>;
    clocks = <CLK_mspi0>;
    interrupts = <GIC_SPI INT_IRQ_MSPI_0 IRQ_TYPE_LEVEL_HIGH>;
    /*
    * padmux:
    * mode=1: PAD_PM_SD_CDZ, PAD_SD_D1, PAD_SD_D0, PAD_SD_CLK, PAD_SD_CMD
    * mode=2: PAD_TTL16, PAD_TTL17, PAD_TTL18, PAD_TTL19
    * mode=3: PAD_GPIO4, PAD_GPIO5, PAD_GPIO6, PAD_GPIO7
    * mode=4: PAD_FUART_RX, PAD_FUART_TX, PAD_FUART_CTS, PAD_FUART_RTS
    * mode=5: PAD_GPIO8, PAD_GPIO9, PAD_GPIO10, PAD_GPIO11
    * mode=6: PAD_GPIO0, PAD_GPIO1, PAD_GPIO2, PAD_GPIO3
    */
    spi0_mode = <5>;
    status = "ok";
};
```

- 需要注意spi用到的这几根pin不能在padmux里面配置成其他用途
- 修改makefile，编译spi 点屏demo
 - sdk\verify\feature\disp\config.mk

```

DEPS:=disp/common

ifeq ($(USE_CASE),fb)
    →SRCS:= fb_ut.c
    →DEPS+=disp/ugui
else
    →ifeq ($(USE_CASE),disp)
    →SRCS:= spi_panel.c
    →else
    →→ifeq ($(USE_CASE),vdec_disp)
    →→SRCS:= vdec_ut.c
    →→endif
    →endif
endif

```

b. project/release/customer_tailor/nvr_i2m_display_glibc_tailor.mk

```

15 interface_venc:=enable
16 interface_wlan:=enable
17
18 misc_fbdev:=enable
19 #verify_zk_full:=enable
20 verify_disp:=enable
21 #mhal
22 #mhal_aio:=disable
23 mhal_csi:=disable
24 #mhal_disp:=disable
25 #mhal_divp:=disable

```

4. 编译spi点屏demo
 - a. cd sdk/verify/feature
 - b. make clean;make disp
 编译生成的demo在**sdk/verify/feature/disp/prog**
5. 将prog 和 240 320 rgb565 20.bin放到同一目录下执行./prog即可点亮panel

```

/mnt/P2/spipanel # ./prog
SPI - Open Succeed. Start Init SPI...
spi mode: 0
bits per word: 8
max speed: 60000 KHz (60 MHz)
init panel start
user export gpio:47
user export gpio:48
user export gpio:6
user export gpio:7
gpio:47 direction set ok
gpio:48 direction set ok
gpio:7 direction set ok
gpio:6 direction set ok
input 'q' exit

```

需要注意的是目前demo的pin的配置如下:

```

CS GPIO47
CS1 GPIO48
RST GPIO6
RS GPIO7

```

```

SPI相关pin:
CZ GPIO8
CLK GPIO9
DI GPIO10
DO GPIO11

```

如果跟demo的pin脚有差异需要在sdk\verify\feature\disp\spi_panel.c中如下地方更改pin的配置

```
1  #include "gpio_operation.h"
2
3  #define CS_PANEL_0 (47)
4  #define CS_PANEL_1 (48)
5
6  #define RS_GPIO_PIN (7)
7  #define RST_GPIO_PIN (6)
8
9
10 static void WaitTime(long ms)
11 {
12     int i;
13     for (i = 0; i < ms; i++)
14     {
15         // do nothing
16     }
17 }
```